Zachariah Hansen

Software Engineer | Full Stack Developer

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Experience

DSD Labs - Full-Stack Software Engineer **April 2026 - Present**

- Designed and built the platform's Model Context Protocol (MCP) server from the ground up, including OAuth 2.1 / Azure AD Government authentication and all 83 AI-callable tools across six business domains (tasks, requirements, recruiting, LMS, and GitLab), enabling AI assistants to operate the system programmatically under full RBAC enforcement and an audited trail.
- Architected and shipped an enterprise Microsoft Teams chatbot for a Laravel monolith on Azure GCC High, serving 107 users with 20+ commands across 8 business domains and Azure OpenAI (GPT-4o) intent classification over 25+ intents; secured it with a custom JWT layer validating Bot Framework tokens against OpenID Connect JWKS and an RBAC gate enforcing page-level permissions per command.
- Re-architected the application-wide role-based access control system, authoring the central permission evaluator and per-module scope-resolver registry that serve as the single source of truth for every access decision across web, API, bot, and MCP surfaces.
- Built a GitLab CI/CD integration with webhook-driven sync that links merge requests, pipelines, and commits to work-breakdown tasks, plus an AI observability layer routing every LLM call through one instrumented, version-controlled client for compliance logging.

  Technologies: PHP | Laravel | PostgreSQL | Azure GCC High | Azure AD | Azure OpenAI | Bot Framework | GitLab

Merits AI - Full Stack Software Engineer **Oct. 2025 - Present**

- Led DoD STIG security compliance for a legal eDiscovery platform, raising compliance across 286 controls from 69% to 98% and remediating all CAT I/II findings plus SQL- and command-injection vulnerabilities.
- Re-architected the Azure Functions document pipeline: parallelized OCR 3-5x, added MinHash/LSH deduplication, and built BAML + Azure OpenAI extraction with hybrid pgvector RAG search, cutting AI tokens ~50%.
- Built an end-to-end document-review batch workflow and a custom PostgreSQL migration framework across a multi-tenant Laravel + Python (FastAPI / Azure Functions) codebase.

  Technologies: PHP | Laravel | Python | FastAPI | Azure Functions | PostgreSQL | pgvector | Azure OpenAI

Aabo Home - Software Engineer **July 2025 - Oct. 2025**

- Developed and maintained iSoftPull credit check integration for customer onboarding, enabling equipment selection and Sunlight Financing using PHP/Laravel and PostgreSQL.
- Implemented comprehensive PHPUnit test coverage for the entire credit check and financing process, improving code reliability and reducing bugs.
- Created technical documentation in Notion for team knowledge sharing, improving onboarding efficiency and development standardization.

  Technologies: PHP | Laravel | PostgreSQL

Salt Lake Weekly - Full Stack Developer **April 2025 - June 2025**

- Migrated the application to React Native for cross-platform support and documented the API with Swagger, improving developer experience.

  Technologies: JavaScript | React | React Native | PostgreSQL | BitBucket | Jira

Motorola Solutions - Software Engineering Intern **Jan. 2025 – Mar. 2025**

- Refactored complex C# in the Premier One Records Management Software, clearing 300+ IDE warnings and reaching 98% unit-test coverage.

  Technologies: C# | MSTest | Visual Studio | Azure

Leadership & Activities

Neumont Hackathon - First Place **May 2025**

- Won 1st place out of 10+ teams in a 48-hour competition, building a machine learning pipeline to predict autism spectrum disorder from questionnaire and demographic data on a heavily imbalanced dataset.
- Benchmarked Logistic Regression, Random Forest, XGBoost, CatBoost, and Neural Networks, using Optuna for hyperparameter optimization to handle class imbalance and maximize predictive performance.

Education

Neumont University **Graduated Sept. 2025**

Bachelor of Science, Software Engineering. GPA: 3.99 Salt Lake City, UT